

Pipeline

[Input Files] → [Detect Source] → [Extract Data] → [Normalize] → [Merge] → [Confidence] → [Projection] → [Validation] → [Final JSON]

Canonical Schema Candidate Information: ID, Name, Email, Phone Location Professional Links: LinkedIn, GitHub, Portfolio Profile Details: Headline(Recruiter Notes), Years of Experience Skills Work Experience: Company, Title, Duration, Summary Education: Institution, Degree, Field, Graduation Year Provenance: Field, Source, Method Confidence Score: Overall profile confidence	Data Normalization • Phone: Convert to E.164 format • Date: Standardize to YYYY-MM • Skills: Convert aliases to canonical names • Country: Standard country name • Email: Lowercase • Name: Title Cased
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Merge / Conflict-Resolution Policy Match key:

- Match using Email first.
- If email missing, use Phone(Never only Name).
- Ignore empty values.
- Remove duplicate skills.
- Keep only one final candidate profile(Canonical Profile).
- A false merge is treated as worse than two correctly-split records
- The higher per-source confidence wins; on an exact confidence tie, SOURCE_PRIORITY breaks it (ats_json > csv > resume > github).
- Same-institution but different degree levels (B.Tech vs M.Tech) are kept as separate entries, not collapsed.
- Confidence depends on: Number of fields available, Agreement between sources and Data completeness.

Runtime Custom Output :

- **custom_config.json decides: Fields to include, Field order and Output format.**
- **Projection:** Implemented by reading config.json, resolving field paths, applying optional normalization, and writing a new projected JSON object.
- **Validation:** Implemented using configuration checks (validate_config) and per-candidate required-field validation with ProjectionError handling.

Edge Cases Handled

- **Garbage/Malformed CSV Row** – Creates a partial profile with low confidence instead of crashing or generating incorrect/invented data.
- **Different Phone Formats** – Phone numbers in different formats (e.g., 9811023456 and +919811023456) are normalized to the same E.164 format and merged correctly.
- **Same Name, Different Person** – Candidates with the same name but no shared email/phone (e.g., namesake resume)/different contact info remain separate records; no fuzzy name matching is used.
- **Missing/Invalid Input Files** – Missing/Invalid Files are skipped with a warning, while the remaining sources continue processing.
- **Duplicate Candidate Records** – Duplicate records from multiple sources are merged into a single canonical candidate profile.

Deliberately descoped under time pressure:

OCR for scanned resumes – Supports only text-based PDF/DOCX resumes.

Live LinkedIn data extraction – LinkedIn has no public API for profile data and scraping violates its Terms of Service. The project only preserves LinkedIn URLs already available in the input data.

Fuzzy/Probabilistic name matching – Candidate matching uses exact identifiers (email, phone, etc.) instead of similarity algorithms such as probabilistic matching.